

Sysinternals — Full Cheat Sheet

What is Sysinternals? A collection of Windows utilities for system administration, troubleshooting, Windows internals, incident response, digital forensics, threat hunting, and malware analysis.

Big idea: Sysinternals gives you deeper visibility into what Windows is doing than many built-in tools. Some legitimate Sysinternals tools can also be abused by attackers (Living off the Land).

Category	Tool	What it does	Run / useful command
File & Disk	Sigcheck	Checks file version, timestamps, digital signatures, certificates and optionally VirusTotal status. Useful for finding suspicious/unsigned executables.	<code>sigcheck.exe file.exe</code> <code>sigcheck.exe -u -e C:\Windows\System32</code> -u → show unsigned/unknown or detected files depending on VT setting. -e → executable images only.
File & Disk	Streams	Finds NTFS Alternate Data Streams (ADS). ADS lets a file have additional named data streams. It is legitimate NTFS functionality but can be abused to hide data.	<code>streams.exe file.txt</code> Read a named ADS in PowerShell: <code>Get-Content .\file.txt -Stream ads.txt</code> Example: <code>file.txt:secret:\$DATA</code>
File & Disk	SDelete	Secure deletion utility. Can delete files/directories and cleanse free space. Attackers may abuse it for indicator removal or data destruction.	<code>sdelete.exe file.txt</code> Warning: deletion is destructive; use only in a lab/when intended.
Networking	TCPView	Shows TCP/UDP endpoints, local and remote addresses, ports, connection states, and the process owning the endpoint. Great for network investigation.	<code>tcpview.exe</code> CLI version: <code>tcpvcon.exe</code> Look at Remote Address + State + Process.
Networking	Resource Monitor*	Built-in Windows utility (not Sysinternals) that can show TCP connections and remote addresses.	<code>resmon</code> Then: Network → TCP Connections
Process	Autoruns	Shows programs configured to run automatically at boot/login and many other auto-start locations. Very useful for finding Persistence.	<code>autoruns.exe</code> Check Everything, Logon, Services, Drivers, Scheduled Tasks, Image Hijacks, etc.
Process	Process Explorer	Advanced Task Manager. Shows processes, PIDs, users, services, DLLs, handles, network activity and digital signatures. Useful for deep process investigation.	<code>procexp.exe</code> Right-click a process → Properties. Use Verify Signatures; inspect Services/TCP-IP/Handles/DLLs.
Process	Procmon	Real-time monitoring of file system, Registry, process and thread activity. Captures huge numbers of events, so filtering is essential.	<code>procmon.exe</code> Ctrl+E → start/stop capture. Ctrl+L → Filter. Filter by Process Name or PID.
Process	ProcDump	Creates process dumps, especially useful for CPU spikes, crashes, troubleshooting and process/memory investigation.	<code>procdump.exe -ma PID dump.dmp</code> Example: <code>procdump.exe -ma 3888 notepad.dmp</code>
Process	PSEXEC	Executes processes/commands on remote Windows systems. Legitimate admin tool but commonly associated with lateral movement and remote execution by attackers.	<code>psexec.exe \\HOST cmd</code> Requires appropriate permissions/network access. MITRE associations in the room: T1570, T1021.002, T1569.002.
Security	Sysmon	Background Windows service/driver that records detailed security-relevant telemetry to Windows Event Logs. Important for SOC detection and SIEM.	Install: <code>sysmon64.exe -accepteula -i</code> Events: Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon Common telemetry includes process creation and network connections; exact events depend on configuration.
System Information	WinObj	Displays the Windows Object Manager namespace: sessions, devices, DosDevices, WindowStations and other internal objects. Useful for Windows internals.	<code>winobj.exe</code> Use it to explore Session 0/1 and Windows object namespaces.
Miscellaneous	BgInfo	Displays system information such as computer name, IP address and OS details on the desktop wallpaper. Handy when managing many servers over RDP.	<code>bginfo.exe</code>
Miscellaneous	RegJump	Takes a Registry path and opens Regedit directly at that location, avoiding manual navigation.	<code>regjump.exe HKLM\SOFTWARE\Microsoft</code>
Miscellaneous	Strings	Extracts readable ASCII/Unicode strings from files, including binaries. Useful for quick malware triage and finding URLs, paths, commands or PDB references.	<code>strings.exe ZoomIt.exe</code> Search: <code>strings.exe ZoomIt.exe findstr /i ".pdb"</code>

Core Notes to Remember

1. File & Disk Utilities

Sigcheck: Think **signature/file verification**. An unsigned executable is not automatically malicious, but it is worth investigating when unexpected.

Streams / ADS: NTFS files have a main data stream and can have named alternate streams. Example: `file.txt:secret`. ADS is not inherently malicious; Windows can use streams such as `Zone.Identifier` for downloaded-file metadata.

SDelete: Think **secure deletion / evidence removal**. It has legitimate administrative uses but can be abused by adversaries.

2. Networking Utilities

TCPView: Think **who is connecting to whom?** Check Process, Local Address, Remote Address, Port and State. An Established connection means a TCP connection is currently established.

WHOIS / reputation: Take the Remote IP and investigate ownership/reputation. Organization/ISP gives context, but ownership alone does not prove a connection is safe or malicious.

Resource Monitor: Built into Windows; use `resmon` → Network → TCP Connections.

3. Process Utilities

Autoruns: Think **Persistence**. It checks Startup, Run/RunOnce, Services, Drivers, Scheduled Tasks, Winlogon, Image Hijacks and other auto-start locations.

Process Explorer: Think **deep process investigation**. Top pane = processes; bottom pane can show handles or DLLs. Useful properties include Services, TCP/IP, Handles and digital signature verification.

Procmon: Think **what is this process doing right now?** It records file-system, Registry, process and thread activity. Use filters aggressively because it can generate thousands of events.

ProcDump: Think **capture process state**. A dump can later be analyzed for crashes, performance problems or suspicious process behavior.

PsExec: Think **remote execution / lateral movement**. It is legitimate administration software, but attackers can abuse it.

4. Security Utilities

Sysmon: Think **security telemetry**. It stays resident across reboots, monitors configured activity, and writes events to Windows Event Logs. Those events can be collected by Windows Event Collection or SIEM agents for detection and investigation.

SOC flow: Windows → Sysmon → Event Logs → SIEM → Detection/Alert → Analyst investigation.

5. System Information

WinObj: Think **Windows internals / Object Manager**. It displays the NT Object Manager namespace. The room uses it to illustrate Session 0 (system/services) and Session 1 (first logged-in user).

Sessions: There can be separate `csrss.exe` instances for different sessions. Session context matters when investigating Windows processes.

Startup relationship: SMSS initializes sessions and starts important processes such as CSRSS/Winlogon as part of session initialization.

6. Miscellaneous

BgInfo: Think **server identification**. Useful when RDPing into many servers.

RegJump: Think **fast Registry navigation**.

Strings: Think **quick binary triage**. It can reveal readable text such as URLs, file paths, commands and PDB paths.

Important Comparisons

Process Explorer vs Procmon: Process Explorer asks “*What is this process and what is associated with it?*” Procmon asks “*What is the process doing right now?*”

TCPView vs Process Explorer: TCPView focuses on network endpoints/connections; Process Explorer gives a broader process investigation view that can include network, services, DLLs and handles.

Sysmon vs Procmon: Procmon is interactive real-time troubleshooting/investigation; Sysmon is persistent background security telemetry written to Event Logs for later analysis/SIEM.

SOC / Blue Team Mental Map

Suspicious Windows Machine

→ **Autoruns:** persistence?

→ **Process Explorer:** what is the process, who owns it, DLLs/handles/services/network/signature?

→ **Procmon:** what files/Registry/process operations is it performing?

→ **TCPView:** where is it connecting?

→ **Sigcheck / Strings:** triage the file

→ **Sysmon + SIEM**: collect telemetry and build detections.

Quick Memory Lines

Sigcheck → Signature / file check

Streams → ADS

SDelete → Secure deletion

TCPView → Network connections

Autoruns → Persistence

Process Explorer → Deep process investigation

Procmon → Real-time activity

ProcDump → Process dump

PsExec → Remote execution / lateral movement

Sysmon → Security telemetry / logging / SIEM

WinObj → Windows Object Manager / internals

BgInfo → System info on desktop

RegJump → Jump to Registry path

Strings → Readable strings from binaries

TryHackMe Lab Path Used in the Room

Many tools in the lab were under:

```
C:\Tools\sysint
```

If a command is not recognized, use the full path, e.g.:

```
C:\Tools\sysint\Procmon.exe
```

The Sysinternals Live UNC path is normally:

```
\\live.sysinternals.com\tools
```

but the TryHackMe lab may be isolated from the Internet, so locally installed tools are often the reliable option.